Tadugam Sai Nithin Reddy

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Introduction

Third-year Computer Science undergraduate with proficiency in Python, C, C++, Java, JavaScript, SQL, and web development, with a strong foundation in Data Structures, Object-Oriented Programming, and Machine Learning. Experienced in developing AI-powered and full-stack applications using React, FastAPI, Node.js, MongoDB, and modern APIs, with a focus on building practical software solutions.

Education

- **Chaitanya Bharathi Institute of Technology, Hyderabad** *2024–2028*
  B.E./B.Tech. – Computer Science and Engineering CGPA: 9.06
- **Narayana Institutions, Hyderabad** *2022–2024*
  Board of Intermediate Education, Telangana CGPA: 9.41

Technical Skills

Programming Languages: Python, C, C++, Java, JavaScript, SQL

Web Technologies: HTML, CSS, React, Node.js, FastAPI

Databases: MySQL, MongoDB, PostgreSQL

Machine Learning: Scikit-learn, XGBoost, Pandas, NumPy

Tools: Git, GitHub, Docker, VS Code

Core Concepts: Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Projects

- **TRAFFIQ – AI Traffic & Accident Risk Prediction System** *Sep 2026*
  React, FastAPI, Python, MongoDB, Scikit-learn
  - Developed an **AI-powered traffic platform** to predict **traffic congestion and accident-risk zones** using historical and real-time traffic data.
  - Built an interactive **React dashboard** with maps, traffic heatmaps, weather information, forecasts, and risk-zone analytics.
  - Implemented **FastAPI REST APIs** and **MongoDB** for traffic-data storage, processing, and retrieval.
  - Integrated **machine learning models** for traffic prediction and accident-risk classification.
- **SureStep – AI-Powered Hiking Safety Platform** *Sep 2026*
  React Native, TypeScript, Node.js, Python, FastAPI, Scikit-learn
  - Developed an **AI-powered hiking safety platform** integrating terrain analysis, GPS navigation, camera-based safety assessment, weather intelligence, and geological hazard monitoring.
  - Implemented a **Machine Learning-based Random Forest model** to analyze terrain and environmental data and predict **risk scores, risk levels, and key risk factors**.
  - Built a **full-stack ML architecture** using React Native, Node.js, and FastAPI, with an AI-powered safety assistant for contextual hiking guidance and real-time risk assessment.
- **TAPAS – Human Thermal Stress & Heat Risk Platform** *Feb 2026*
  React, Python, FastAPI, MongoDB, Machine Learning
  - Developed a **heat-risk platform** integrating temperature, humidity, wind, and radiation data to calculate the **Human Thermal Stress Index (HTSI)** and identify vulnerable populations.
  - Implemented **heat-risk visualization, alerts, cooling locations, worker-risk analysis, and healthcare monitoring** to support community-level response to extreme heat.

Certifications & Achievements

- **React Web Development Certification**
- **Generative AI Certification**
- **Java Programming Certification**
- **Secured Rank 2337 in TS EAMCET 2024**

Positions of Responsibility

- **Ramanujan Mathematics Club (RMC)** – Public Relations Team
- **Chaitanya Spandana** – Social Welfare Club Member